

▼ Handset Connectivity

You don't necessarily need data cards to be mobile. You could always use a regular phone-to-laptop data cable, or Bluetooth, and connect via GPRS or CDMA.

GPRS (GSM's wireless data technology package) plans cost as little as Rs 150 per month in some post-paid schemes, going up to approximately Rs 600.

Though real-life speeds for GPRS are reported to be just 56 to 96 kbps, costs are generally not linked to usage time or data transfer, so this works fine for the casual surfer.

Most CDMA (Code Division Multiple Access) handsets have an inbuilt modem, so once it's connected to your laptop, you can dial up and get online. Some users have reported download speeds for CDMA handsets to be about 120 kbps.

where" broadband access. But what options do we have until that rosy picture materialises?

The Mobile Revolution

With the Government delicensing some frequencies in the 2.4 GHz and 5.1 GHz range, the number of public access hot spots are set to mushroom. Meanwhile protocols such as WAP, GPRS, EDGE, and CDMA offer mobile access solutions.

Typically, we access the Internet over these technologies by connecting our phones to our desktops or laptops and dialling up. This can often be a frustrating experience—for example, what if your phone's battery goes dead in the middle of a browsing session? If your phone isn't Bluetooth-enabled, you've got to deal with a mess of cables. And do you really want your phone balancing precariously on your knee while you're in a train?

Enter The Data Card

Slotted into the standard type II PCMCIA (Personal Computer Memory Card International Association) card slot in notebooks, data cards offer wireless connectivity over the mobile phone network. With a data card, your laptop can be used to make and receive voice calls and SMSes as well. In short, data cards function as modems while also mobile phone-enabling your laptop. They offer you the option of just clicking an icon on your Desktop to get connected.

At one point of time, data cards weren't an affordable option for most people. But that's changed. If there were an equivalent of Moore's law for hardware pricing, data cards would

more than satisfy it. A card that cost about Rs 14,000 to 17,000 in June 2005 now costs about Rs 6,500. Data cards are available from Reliance Infocomm, Airtel, Tata Indicom and Hutch.

Data card vendors point out the cost advantage over Wi-Fi in hotels for the business traveller. But though data cards may lose miserably when it comes to speeds, there's just no competition when it comes to actual mobility. Wi-Fi and WiMax just don't have enough of a presence to compete with data cards.

In addition, service providers claim that data cards will free us from the problems associated with dial-up connections such as time-outs, blackouts and frequent disconnections.

▼ UMTS

Universal Mobile Telecommunications System represents an evolution in terms of capacity, data speeds and new service capabilities from second-generation mobile networks. Using W-CDMA radio access technology that builds on GSM, it provides a clear evolutionary path for more than 80 per cent of the world's wireless market to offer speeds of up to 2 Mbps, and all the benefits of broadband connectivity while on the move. The protocol was evolved by a global alliance of telecommunications standards bodies headquartered in France.

CDMA vs. GSM Data Cards

As against the claimed speeds of 144 to 153 kbps for Tata Indicom and Reliance data cards, both of which use CDMA, Airtel customer support promises 247 kbps for its GSM/EDGE data card. Hutch, the other GSM player, claims just three times the speed of GPRS.

GPRS usually offers lower data bandwidth when compared to CDMA's high-speed technology 1xRTT (short for single carrier radio transmission technology), which has the capability of providing Integrated Services Digital Network (ISDN)-like speeds of as much as 144 kbps. However, 1xRTT requires a dedicated connection to the network for data transfer, whereas GPRS sends packets, which is why data calls made on a GSM handset won't block out voice calls like they do on CDMA phones.

Also, data transmission over CDMA is traditionally considered more secure than GSM, which allows multiple users to share the same channel.

An official from a leading CDMA player wrote off the EDGE (Enhanced Data for Global Evolution) claim of 247 kbps by questioning the number of EDGE-enabled towers. In Mumbai, for example, he said, the speeds at, say Nariman Point and Vashi may differ widely when EDGE "falls back" to GPRS (See box *EDGE To GPRS Fallback*). CDMA, on the other hand, he said, promises a uniform pan-India experience. While admitting that speeds would be inversely proportional to the volume of traffic handled by a particular tower, he stated 70 kbps as the actual average speed one can get with a CDMA data card.

But GSM still has the advantage of global connectivity. Of the 36 countries and 49

▼ Cost And Plan Charges

	One-time Cost	Monthly Charge (Rs)	Data Usage Cap
AirTel	12,999	399	100 MB
		599	1 GB
		799	1.5 GB
Hutch	12,000	499	100 MB
Reliance	6500	650	1 GB
		900	1.5 GB
		1,500	Unlimited
Tata Indicom 9995		700	1 GB